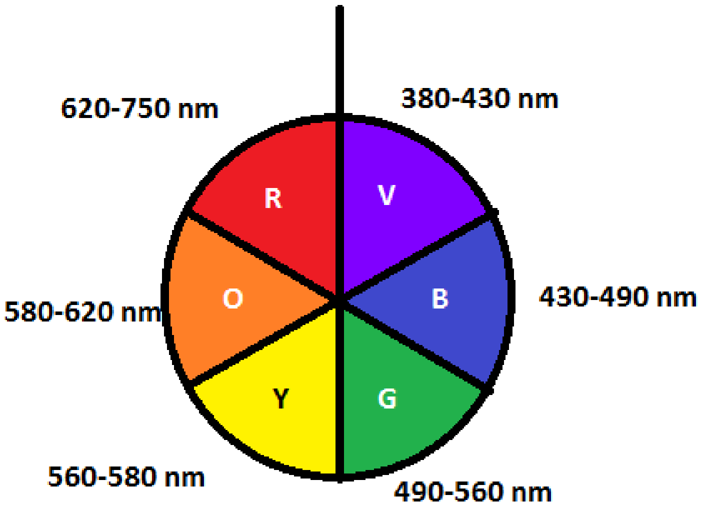
Lecture 8

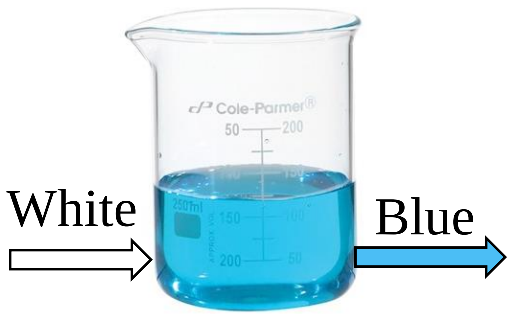
Crystal Field Theory

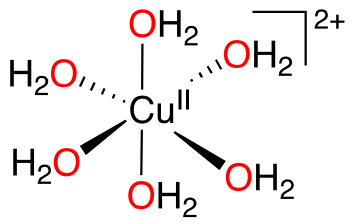
Colors of transition metal complexes



Why we observe color?

Colored solutions contain species that can absorb photons of visible light and use the energy of those photons to promote electrons in the species to higher energy levels. Because the energies of photons are related to the frequencies (and wavelengths) of light (Planck's equation, E = h*v*), only certain wavelength components are absorbed as white light passes through the solution. The emerging light, because it is lacking some wavelength components, is no longer white; it is colored.





Color of light absorbed	Approximate wavelength ranges / nm	Color of light transmitted
Red	700–620	Green
Orange	620–580	Blue
Yellow	580–560	Violet
Green	560–490	Red
Blue	490–430	Orange
Violet	430–380	Yellow

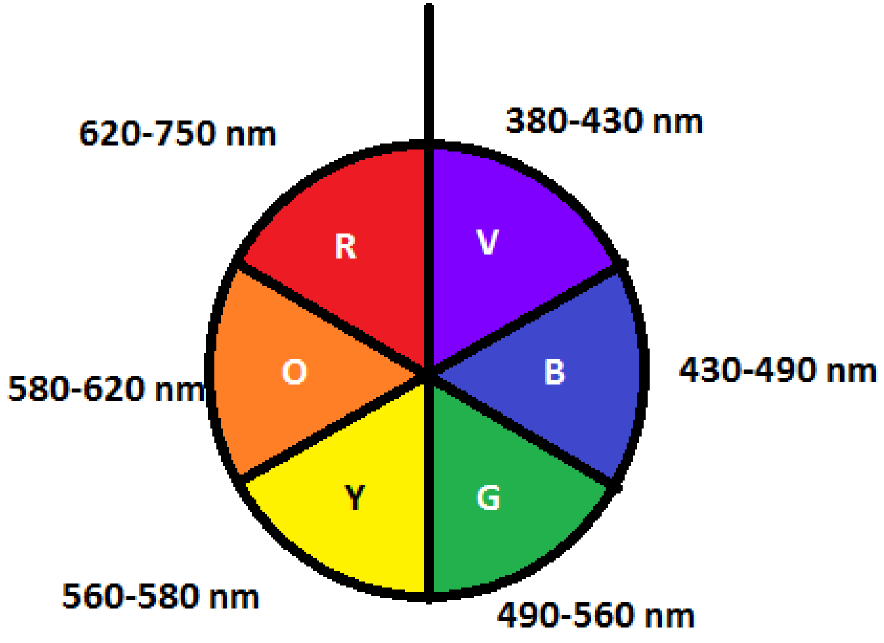
Problem solving!

$[Co(NH_3)_6]^{3+}$ is diamagnetic and orange yellow. $[CoF_6]^{3-}$ on the other hand is paramagnetic and blue . Why ?

$[Co(NH_3)_6]^{3+}$: $t_{2g}^{\,6}$ diamagnetic = no unpaired electrons Orange yellow means absorption in the Violet-blue region (higher frequency, larger Δ_o) d^6 LS

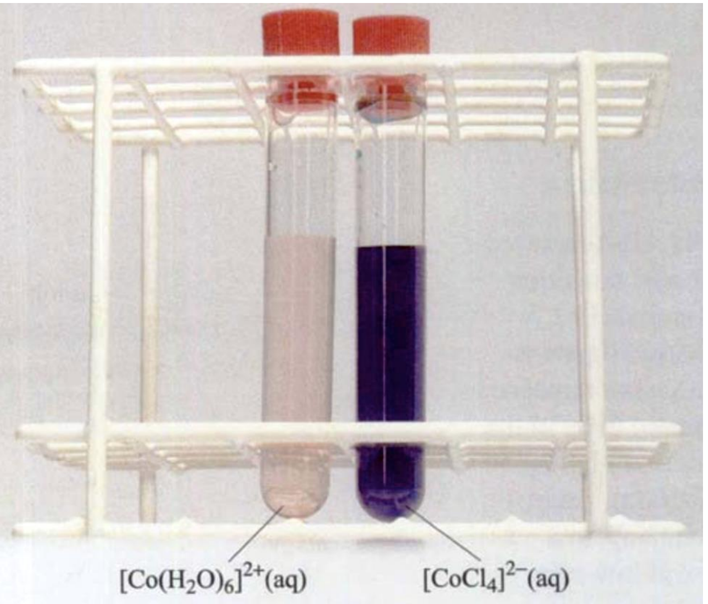
$[CoF_6]^{3-}$ paramagnetic $t_{2g}^{\,4}e_g^{\,2}$ Blue means absorption in the orange region (lower frequency, smaller Δ_o) therefore d^6 HS

Δ_o splitting magnitude is higher for $[Co(NH_3)_6]^{3+}$ due to stronger ligand.

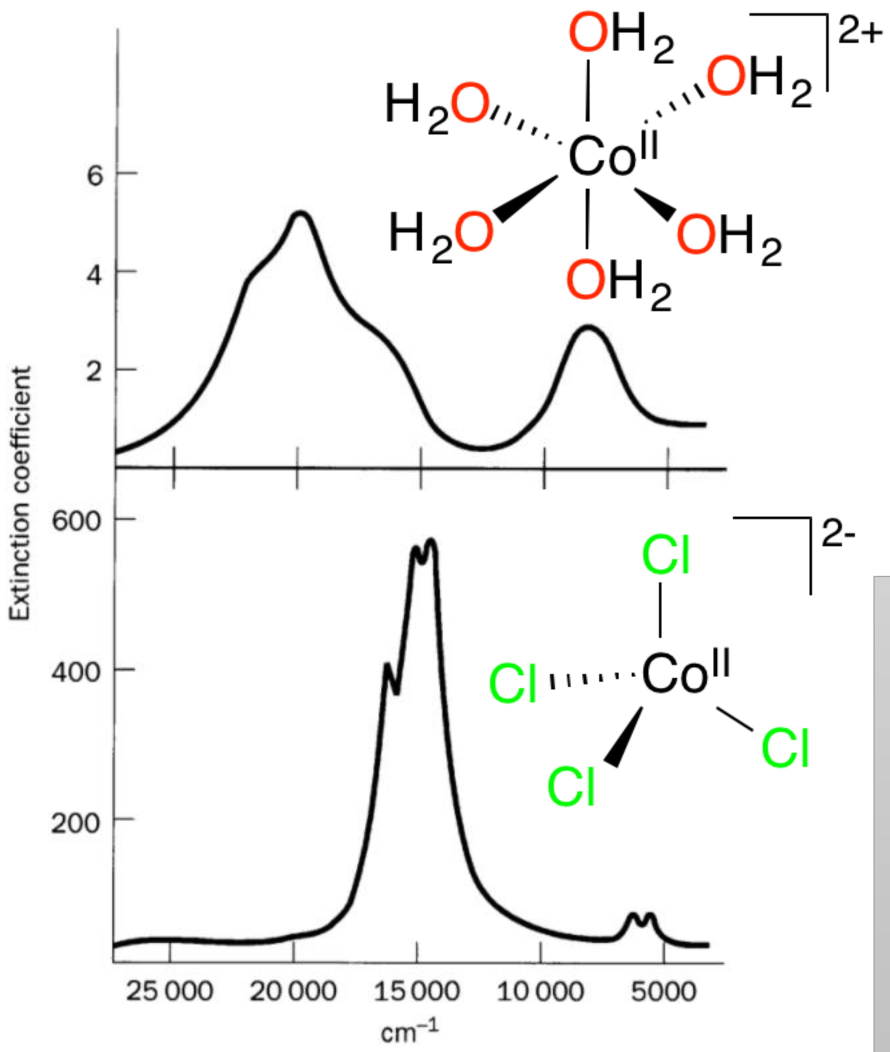


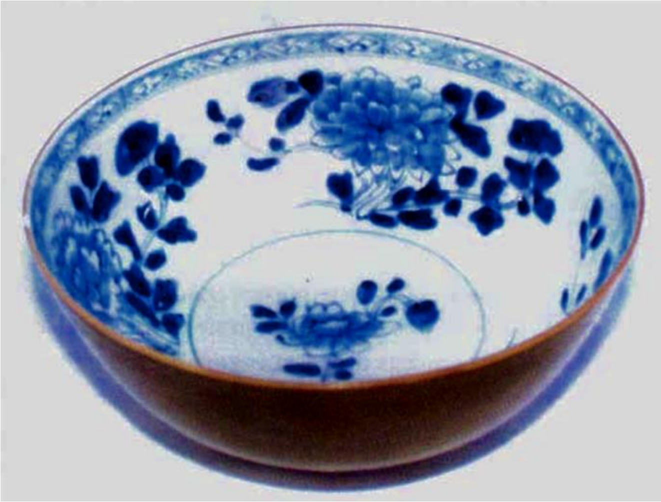
$$[Co^{II}(H_2O)_6]^{2+}(aq) + 4\,Cl^-(aq) \rightleftharpoons [Co^{II}Cl_4]^{2-}(aq) + 6\,H_2O\,(l)$$

pink **blue**



The pale pink solution (left-hand tube) was formed by dissolving cobalt(II)chloride in water. On adding concentrated hydrochloric acid, a deep blue solution (right-hand tube) containing the ion $[CoCl_4]^{2-}$ resulted.





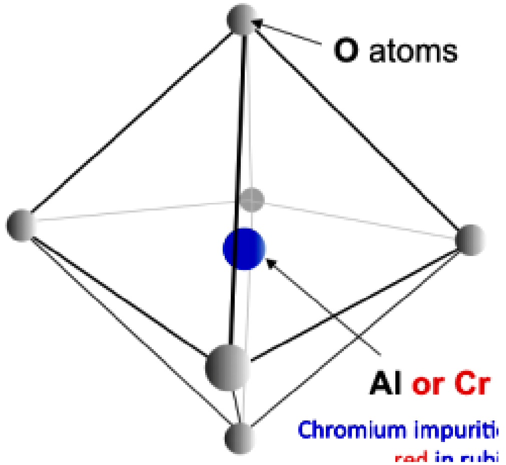
A Chinese bowl from the 18th century, decorated with a cobalt(II) glaze.

The colours of tetrahedral complexes are far more intense than their octahedral counterparts at the same concentration. This is because they do not possess a centre of symmetry, and, consequently, the Laporte selection rule is not strictly applicable.

Origin of color in Gem Stones : Ruby and Emerald

corundum (Al_2O_3) and **Ruby**: corundum with chromium as impurity.

Ruby is Al_2O_3 containing around 0.2—1 atom per cent Cr^{3+} ions in place of the Al^{3+} ions and its red colour results from the absorption of green light in the visible spectrum as a result of the excitation of Cr 3*d* electrons



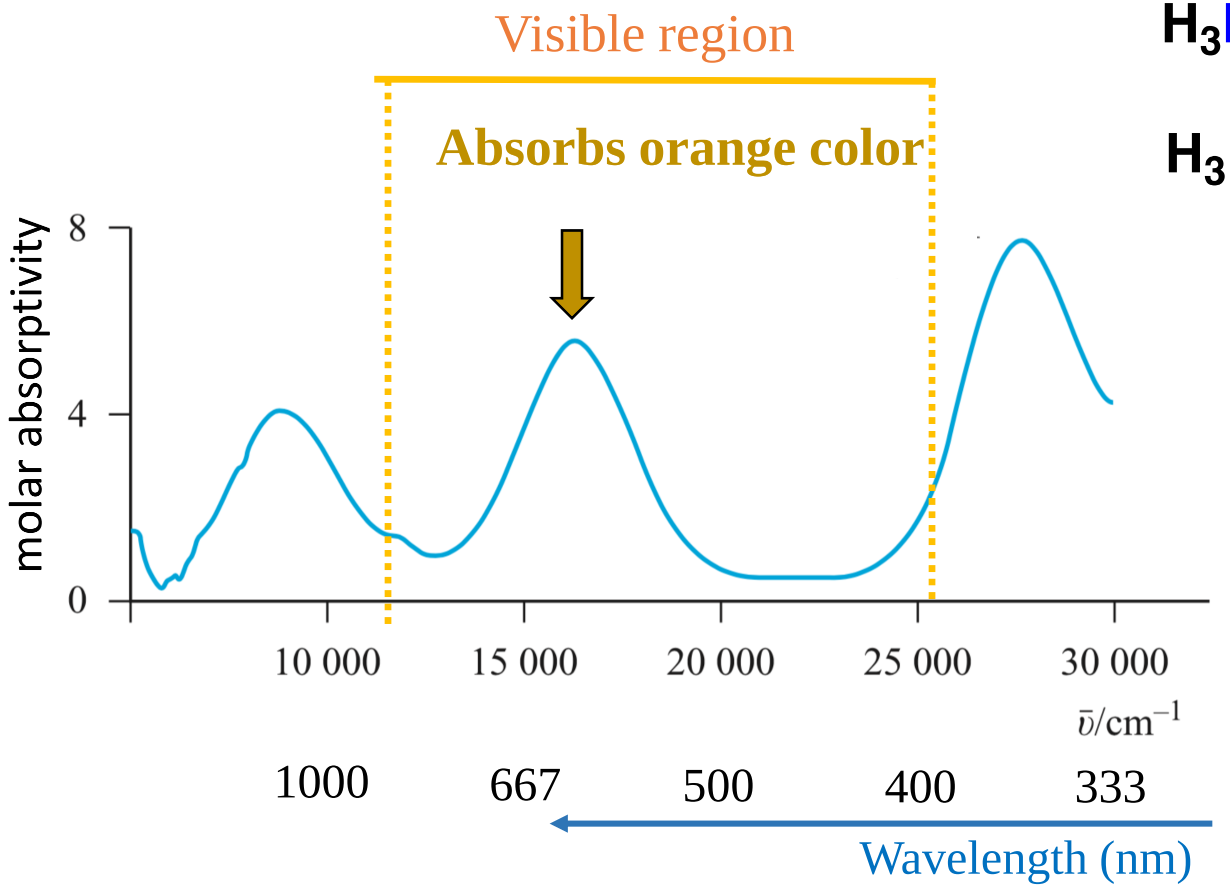
Mineral beryl ($Be_3Al_2(SiO_3)_6$), and **Emerald:** Beryl colored green by trace amounts of chromium and/ or vanadium

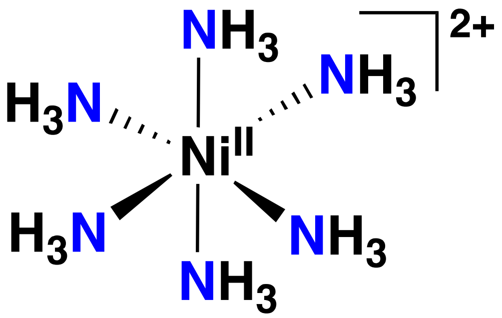
In Emerald, the host structure is beryl, beryllium aluminium silicate, $Be_3Al_2(SiO_3)_6$, and the Cr^{3+} ion is surrounded by six silicate ions, rather than the six O^{2-} ions in ruby, producing absorption at a different energy.

Peridot

Fe^{2+} replacing Mg^{2+} in 6-coordinate sites in Mg_2SiO_4

Electronic spectrum of Ni(II) complex

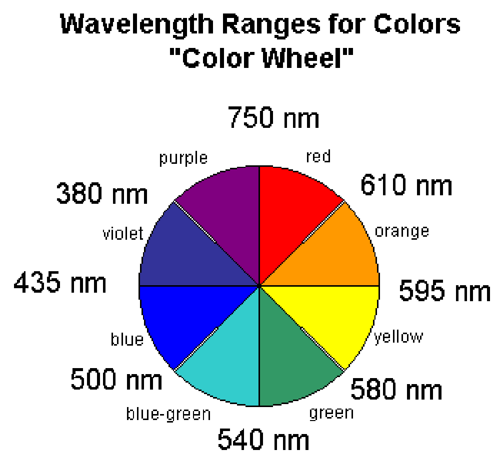


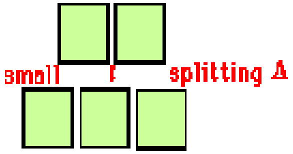


The frequency, wavelength or energy of a transition relates to the energy required to excite an electron:

- **depends on Δ_{oct} and ligand-field strength of the molecule.**
- **decides colour of molecule**

Effects of the metal oxidation state and of ligand strength on color.



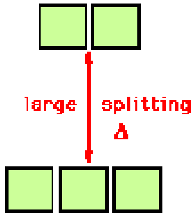


$[V(H_2O)_6]^{2+}$

560 nm
17900 cm^{-1}

$[V(H_2O)_6]^{3+}$

460 nm
21700 cm^{-1}



Intensity of color in a transition metal complex

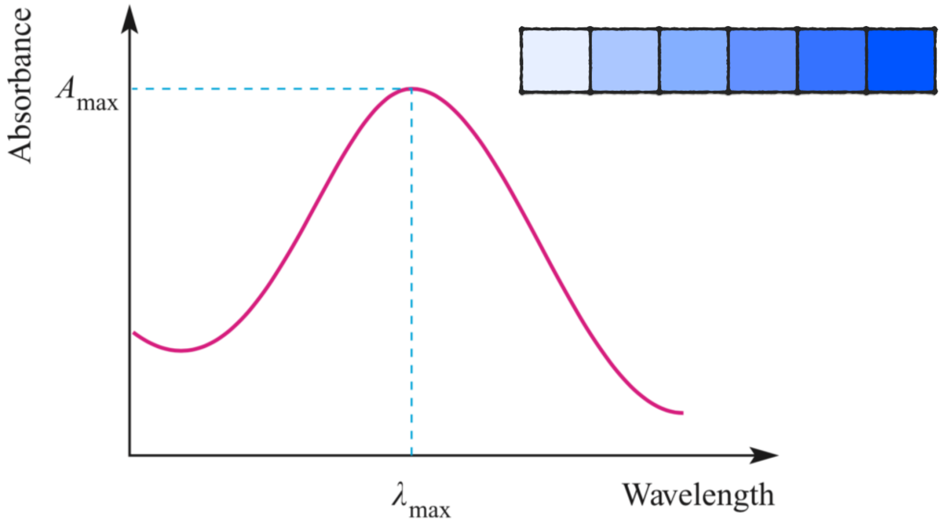
The absorbance is related to the concentration of the solution by the Beer–Lambert law:

$$\mathbf{A} = \boldsymbol{\varepsilon} \times \boldsymbol{c} \times \boldsymbol{l}$$

The **molar absorption coefficient**, **molar extinction coefficient**, or **molar absorptivity** (ε), is a measurement of how strongly a chemical species absorbs light at a given wavelength. It is an intrinsic property of the species; the actual absorbance A, of a sample is dependent on the pathlength, ℓ, and the concentration, c, of the species via the **Beer–Lambert law,**

Selection rules:
Electronic transitions in a complex are governed by Selection rules

A selection rule is a quantum mechanical rule that describes the types of quantum mechanical transitions that are permitted.
They reflect the restrictions imposed on the state changes for an atom or molecule during an electronic transition.
Transitions not permitted by selection rules are said **forbidden**, which means that theoretically they must not occur (but in practice may occur with very low probabilities).



Electronic absorption spectra may be presented as plots of absorbance against wavenumber or wavelength, or as plots of ε_{max} against wavenumber or wavelength.

Laporte Selection Rule

Statement : Only allowed transitions are those occurring with a change in parity (flip in the sign of *one* spatial coordinate.) OR During an electronic transition the azimuthal quantum number can change only by ± 1 $(\Delta\, l = \pm 1)$

Otto Laporte
German American Physicist

Practical meaning of the Laporte rule

Allowed transitions are those which occur between gerade to ungerade or ungerade to gerade orbitals

Allowed

$g \rightarrow u$ & $u \rightarrow g$

Not allowed (FORBIDDEN)

$g \nrightarrow g$ & $u \nrightarrow u$

Gerade = symmetric w r t centre of inversion

Ungerade = antisymmetric w r t centre of inversion

Laporte allowed transitions

$s \leftrightarrow p,\ p \leftrightarrow d,\ d \leftrightarrow f\ (\Delta l = \pm 1)$

Laporte forbidden transitions

$s \leftrightarrow s$, $\mathrm{p} \leftrightarrow \mathrm{p}$, $d \leftrightarrow d$, and $f \leftrightarrow f$ $(\Delta l = 0)$

This rule affects Octahedral and Square planar complexes as they have center of symmetry

Tetrahedral complexes do not have center of symmetry: therefore this rule does not apply

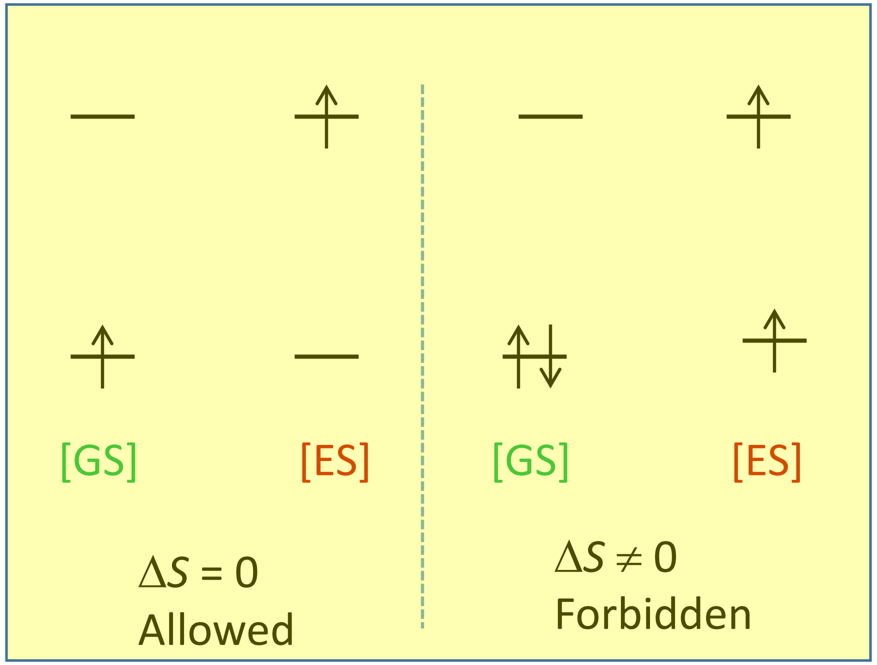
$t_{2g} \rightarrow e_g$ is forbidden or according to Laporte selection rule $d \rightarrow d$ transitions are not allowed !

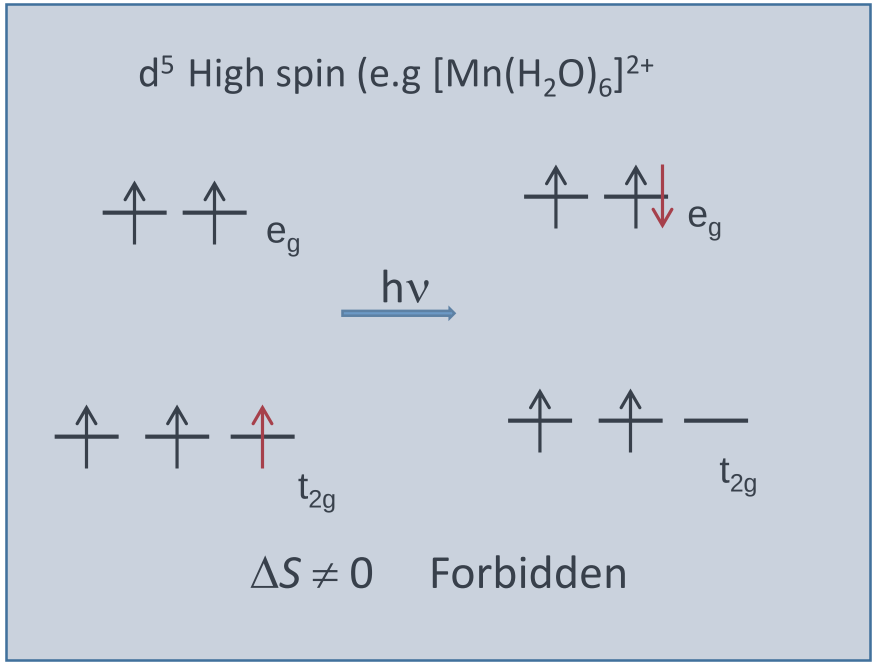
Spin Selection Rule

Statement : This rule states that transitions that involve a change in spin multiplicity are forbidden. According to this rule, any transition for which $\Delta S = 0$ is allowed and $\Delta S \neq 0$ is forbidden

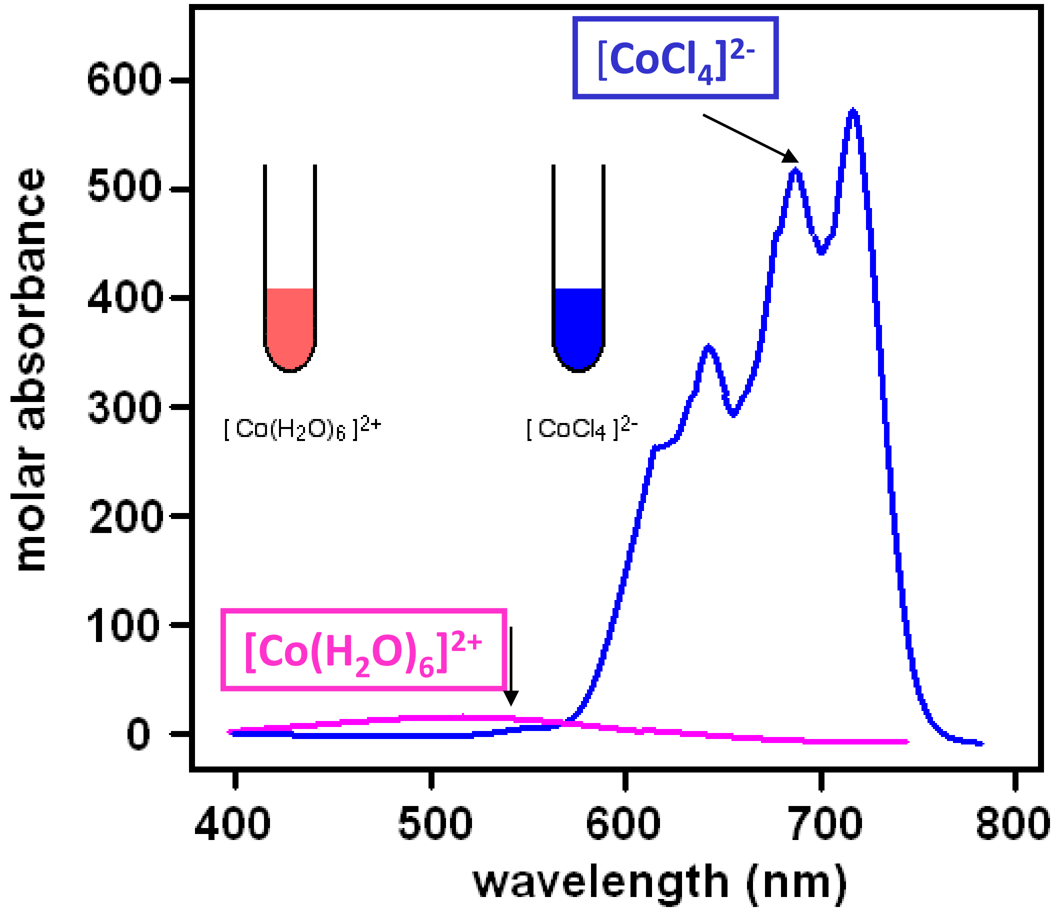
Practical significance of the Spin Selection rule

During an electronic transition, the electron should not change its spin ((An initially antiparallel pair of electrons cannot be converted to a parallel pair, so a singlet (S=0) cannot undergo a transition to a triplet (S=1) and vice versa.

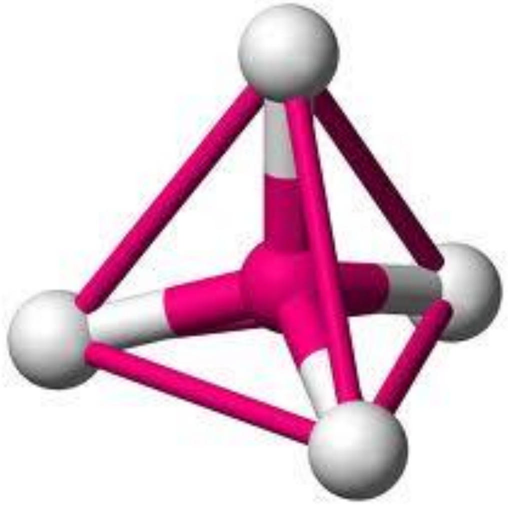




The spectra of octahedral $[Co(H_2O)_6]^{2+}$ and tetrahedral $[CoCl_4]^{2-}$ ions:



Intense d-d bands in the blue tetrahedral complex $[CoCl_4]^{2-}$, as compared with the much weaker band in the pink octahedral complex $[Co(H_2O)_6]^{2+}$. This difference arises because the $\boldsymbol{T_d}$ complex has no center of symmetry, helping to overcome the g→g Laporte selection rule.



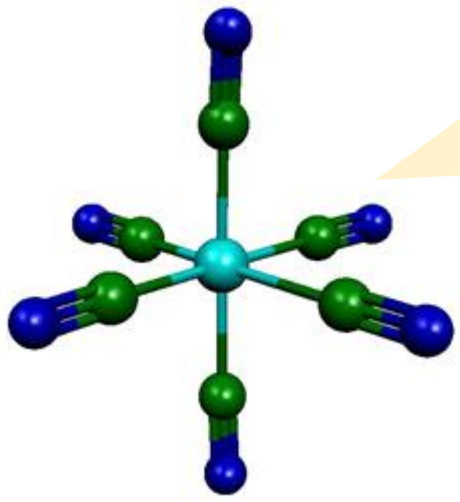
Classification of intensities of electronic transitions

Transition type	Example	Typical values of ε /$m^2 mol^{-1}$	
Spin forbidden, Laporte forbidden (partly allowed by spin–orbit coupling)	$[Mn(H_2O)_6]^{2+}$	0.1	
Spin allowed (octahedral complex), Laporte forbidden (partly allowed by vibronic coupling and d-p mixing)	$[Co(H_2O)_6]^{2+}$	1 - 10	
Spin allowed (tetrahedral complex), Laporte allowed (but still retain some original character)	$[CoCl_4]^{2-}$	50 - 150	 
Spin allowed, Laporte allowed e.g. charge transfer bands	$KMnO_4$	1000 - 10^6	

Relaxing the selection rules

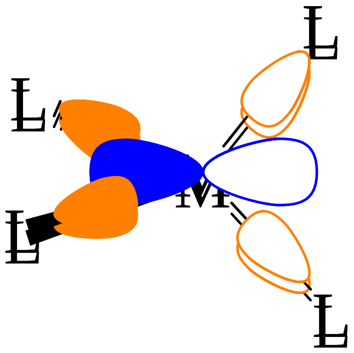
Vibronic coupling

An octahedral complex possesses a centre of symmetry, but molecular vibrations result in its temporary loss, which partially relaxes the Laporte selection rule.

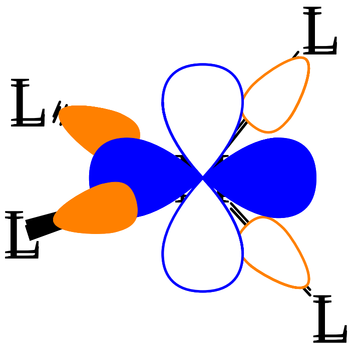


Molecules are not rigid but always vibrating. During this vibration, centre of inversion is temporarily lost

Mixing of states



A metal *p*-orbital overlaps with ligand orbitals



A metal *d*-orbital overlaps with the same ligand orbitals

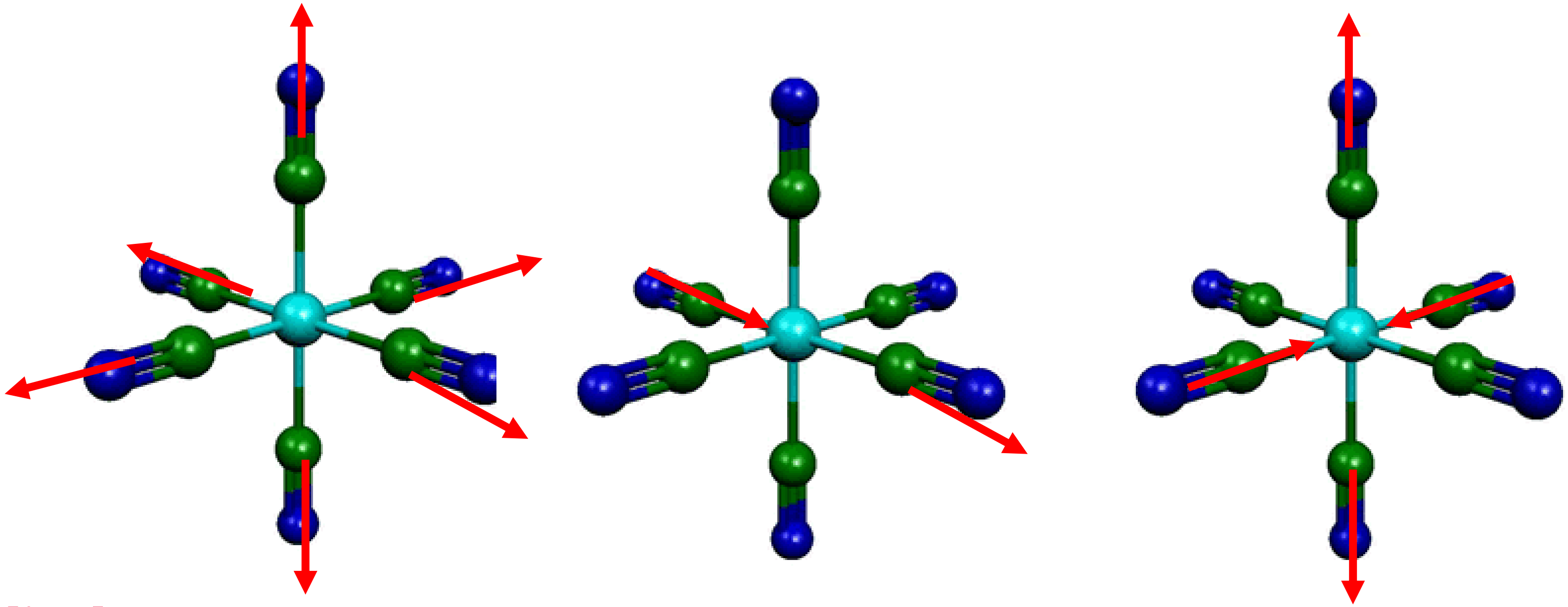
As the molecular orbitals are actually mixtures of *d* and *p*-orbitals, they are partially allowed as $\Delta l = \pm 1$

Spin-orbit coupling

$[Mn(H_2O)_6]^{2+}$

Electrons can have magnetism due to the spin and orbital motions. This coupling allows the spin forbidden transitions to occur. Spin-orbit coupling gets stronger as elements get heavier and so spin forbidden transitions get more important.

During the transition, there must be a change in the dipole moment of the molecule



dipole moment change? | no | yes | no

If there is a large change, the light / molecule interaction is strong and many photons are absorbed:

large area or intense bands → intense colour

If there is a small change, the light / molecule interaction is weak and few photons are absorbed:

low area or weak bands → weak colour

If there is no change, there is no interaction and no photons are absorbed

Charge-transfer bands

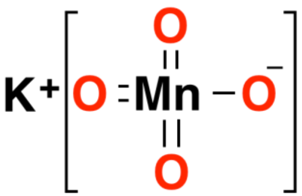
What is a charge-transfer transition?

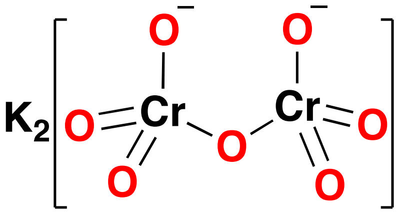
There are many compounds where either the metal ion is highly oxidizing and the ligands are reducing or that the metal ion is highly reducing and the ligands are oxidizing. In such cases there occurs transfer of charge. (i.e electron)from the reducing partner to the oxidizing partner. The metal ion many not possess d-electons.

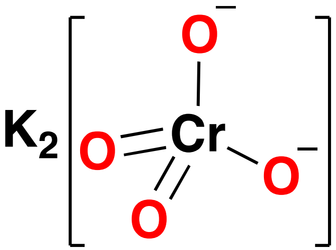
What is "Ligand-To-Metal (LMCT)" Charge Transfer?

Ligand-to-metal charge-transfer transitions are observed in the visible region of the spectrum when the metal is in a high oxidation state and ligands contain nonbonding electrons.

Examples of LMCT transitions

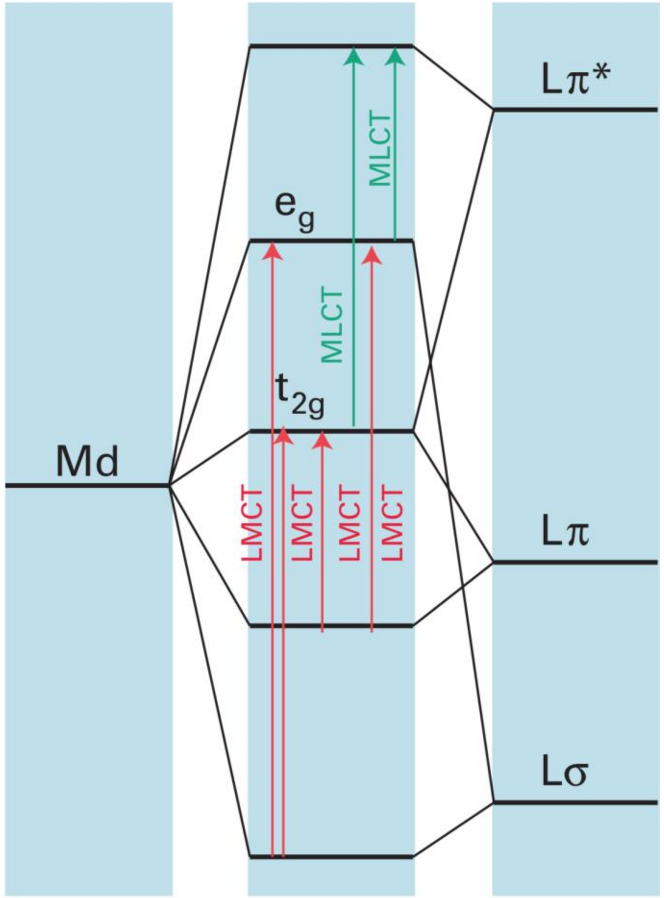






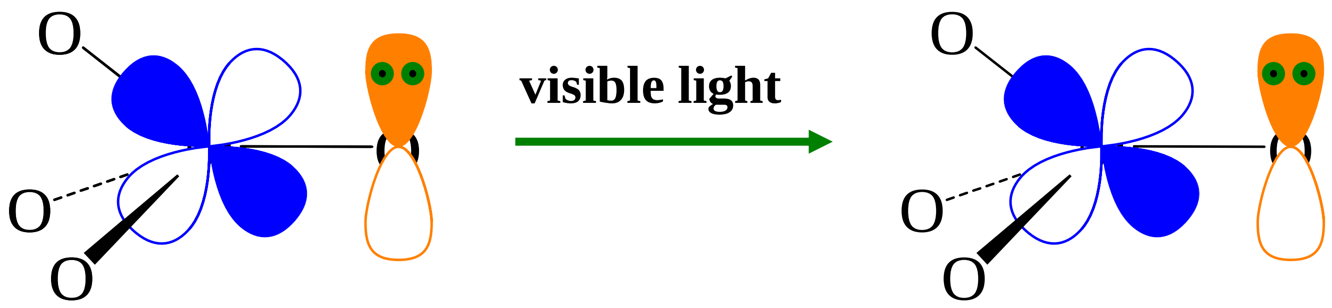
(a)	t_{2g}	$\longleftarrow$	π
(b)	e_g	$\longleftarrow$	π
(c)	t_{2g}	$\longleftarrow$	σ
(d)	e_g	$\longleftarrow$	σ

Possible LMCT transitions



A summary of the charge-transfer transitions in an octahedral complex.

Charge-transfer bands



LMCT transitions are *usually* in the ultraviolet region. They occur in the visible or near-ultraviolet if:

- Metal center is easily reduced. For example when a metal is in high oxidation state.
- Ligand is easily oxidized.

d^0

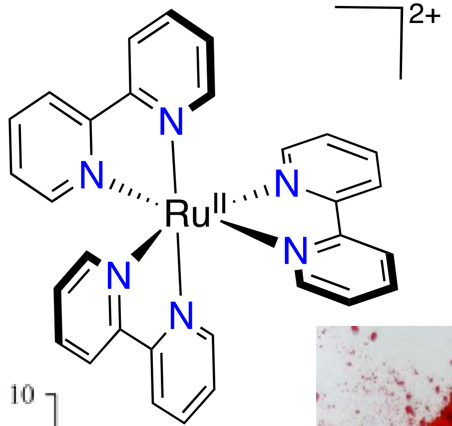
TiO_2	VO_4^{3-}	CrO_4^{2-}	MnO_4^-
Ti^{4+}	V^{5+}	Cr^{6+}	Mn^{7+}
in far UV	~39500 cm^{-1}	~22200 cm^{-1}	~19000 cm^{-1}
white	white	yellow	purple

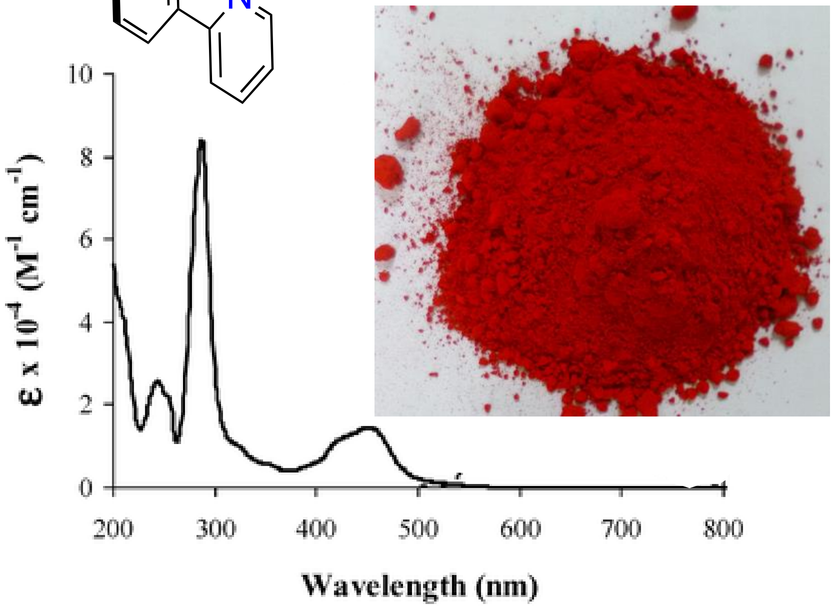
more easily reduced

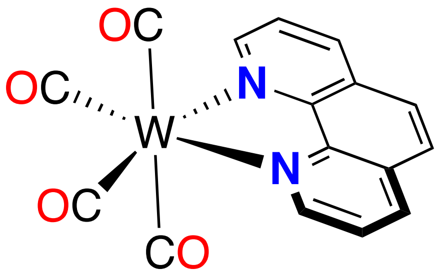
Charge-transfer bands

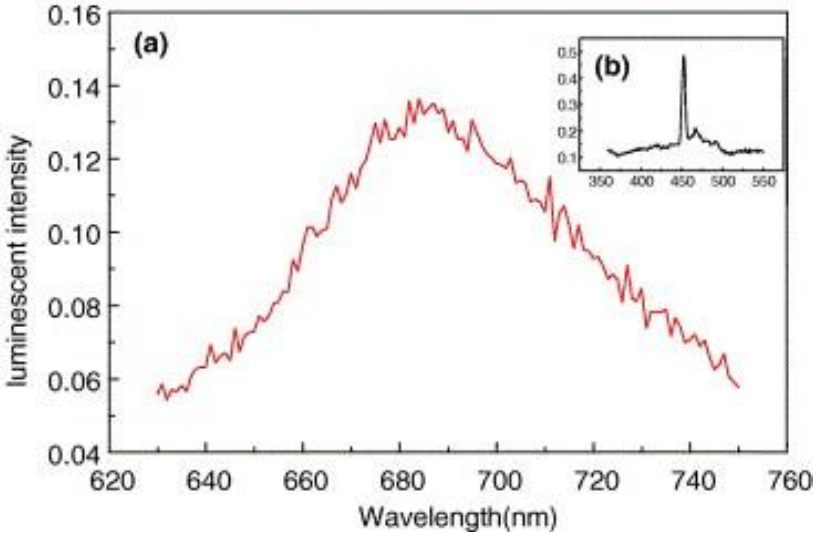
What is “Metal-To-Ligand (MLCT)” Charge Transfer?

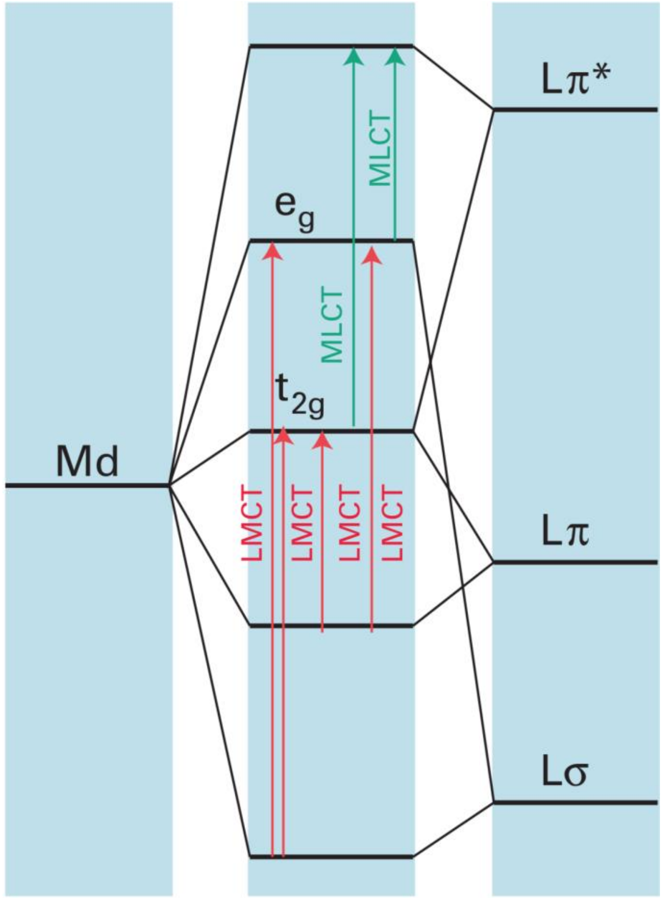
Charge-transfer transitions from metal to ligand are observed when the metal is in a low oxidation state and the ligands have low-lying acceptor orbitals (like π^* orbitals), especially aromatic ligands.







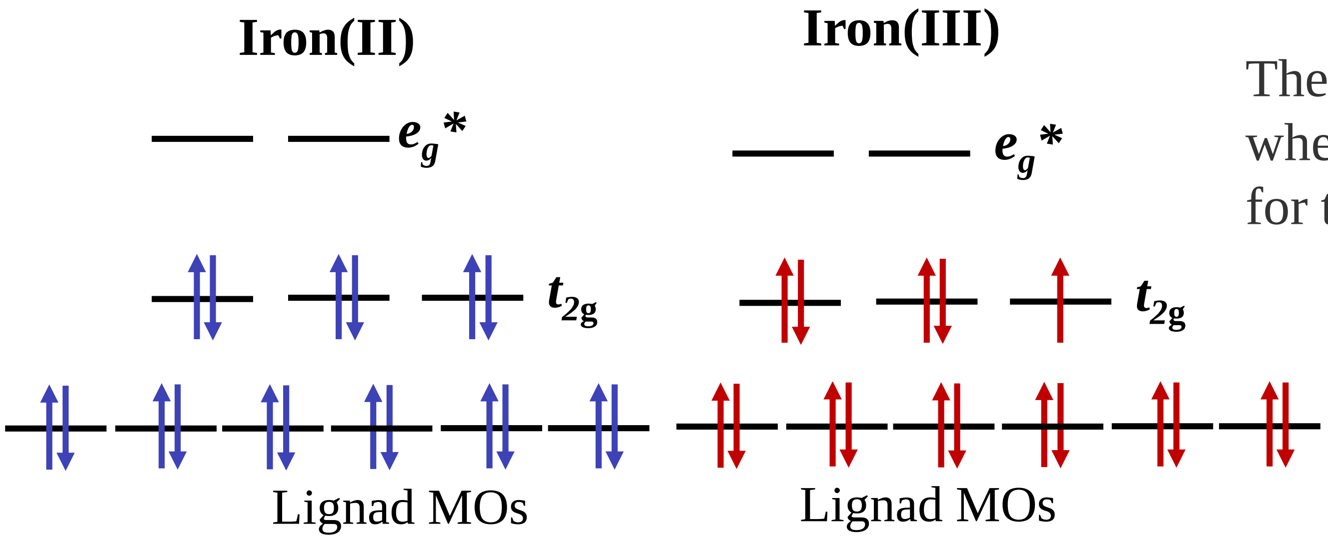


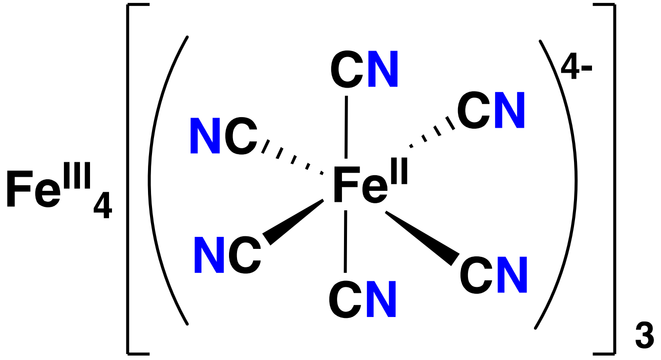


A summary of the charge-transfer transitions in an octahedral complex.

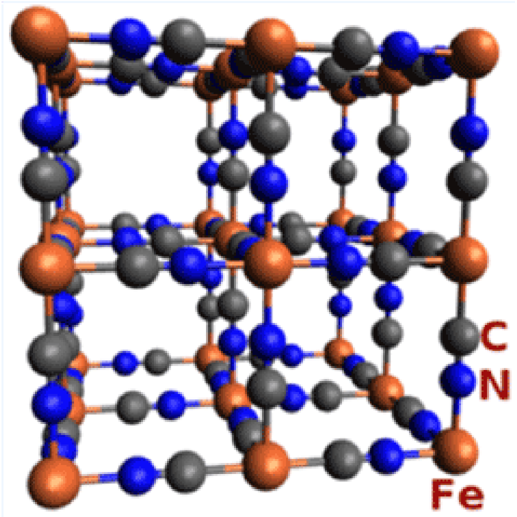
What is "Metal-To-Metal" Charge Transfer?

When there are two or more metal centres close together, spectral transitions can occur between the orbitals based on one metal and those on the other. An interesting set of complexes is those with one metal in two different oxidation states. Such complexes are often intensely coloured.





Prussian blue



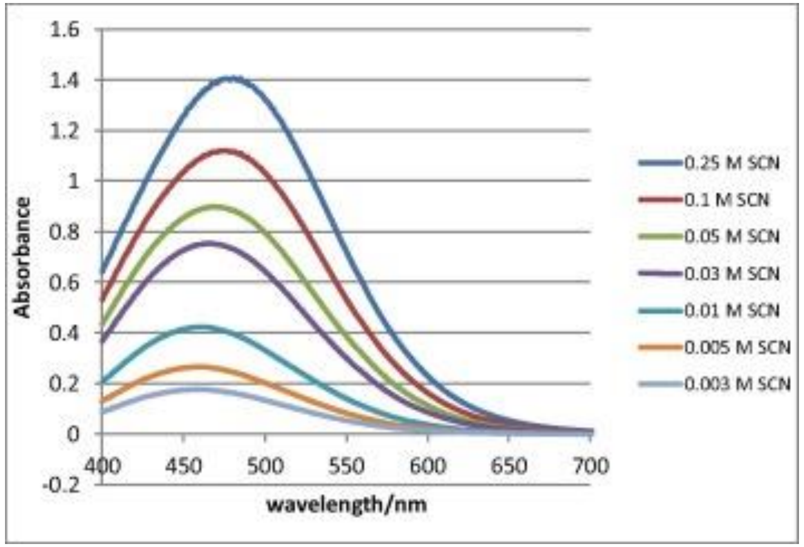
The name Prussian blue originated in the 18th century, when the compound was used to dye the uniform coats for the Prussian army.

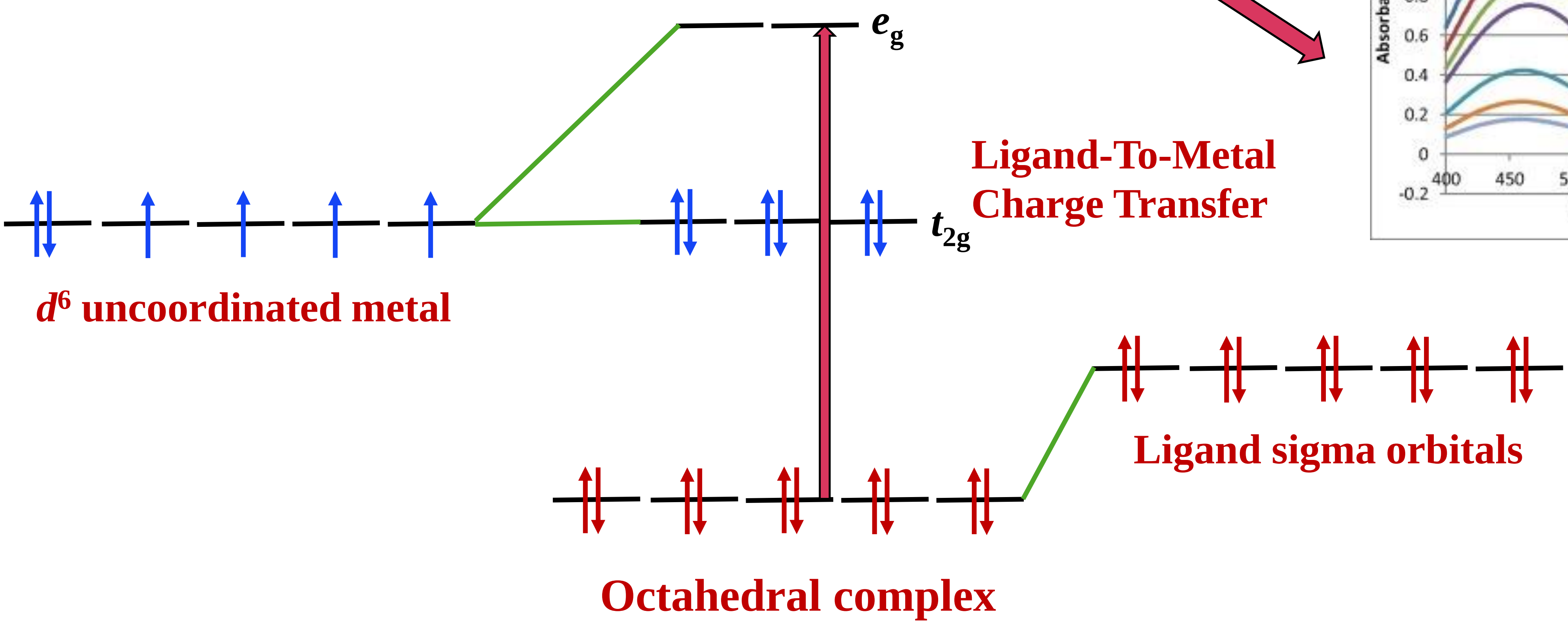
In Prussian blue, $Fe^{III}_4[Fe^{II}(CN)_6]_3$, the iron(III) ions are octahedrally surrounded by the nitrogen atoms of the $[Fe^{II}(CN)_6]^{4-}$ ion. The blue colour is due to transitions from a t_{2g} orbital on iron(II)) in the $[Fe^{II}(CN)_6]^{4-}$ ion to the t_{2g} and e_g* orbitals on iron(III).

Charge-transfer bands

$$[Fe(H_2O)_6]^{3+} + 3\ SCN^- \rightleftharpoons [Fe(SCN)_3(H_2O)_3] + 3\ H_2O$$

pale yellow *colorless* *blood red*





Ligand to Metal Charge Transfer (LMCT) involving an octahedral d^6 complex

Problem Solving

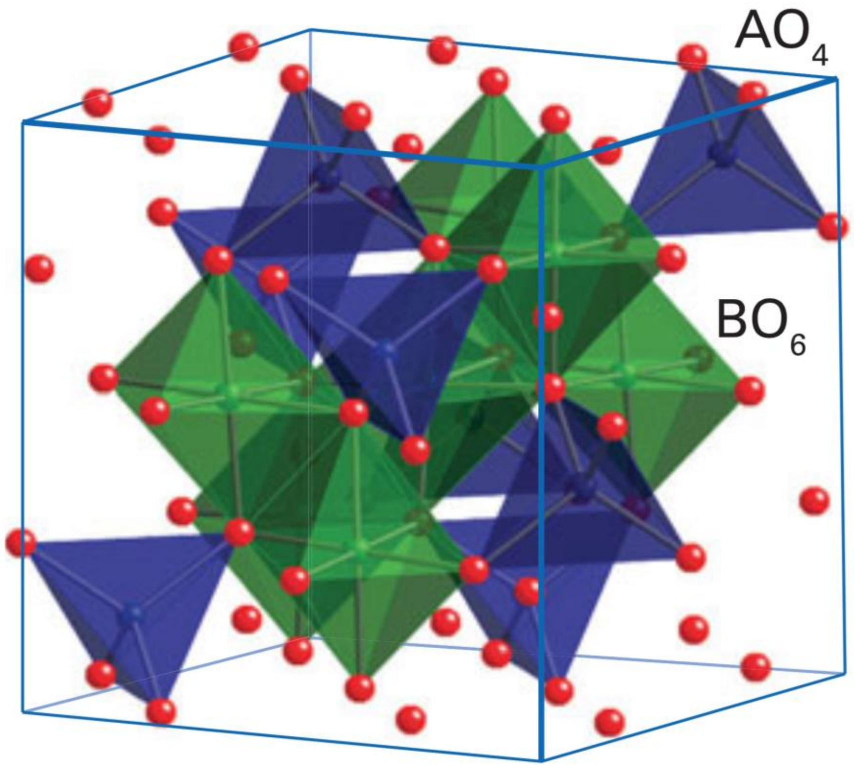
Arrange the given metal complexes in the increasing order of intensity of color (ε) shown by them. Justify your order by writing below each the status of the selection rules for these complexes (no partial marks)

$Fe_4[Fe(CN)_6]_3$, $[CoBr_4]^{2-}$, $[MnF_6]^{4-}$, $[Co(H_2O)_6]^{2+}$

Least intense			Most intense

Spin Forbidden Laporte Forbidden	Spin Allowed Laporte Forbidden	Spin Allowed Laporte Allowed (Tetrahedral)	Spin Allowed Laporte Allowed Charge – Transfer transition

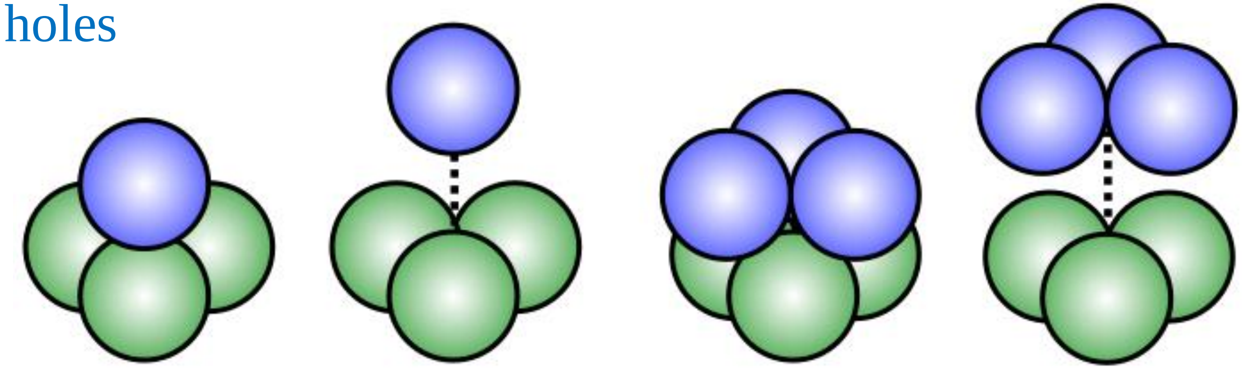
Site preference of Spinels and Inverse Spinels



A segment of the spinel (AB_2O_4) unit cell showing the tetrahedral environment of A ions and the octahedral environments of B ions.

Spinels are a class of crystalline solids of the general formula AB_2O_4 ($A^{II}B^{III}_2O_4$) where

The weak field oxide ions provide a cubic close-packed lattice. In one unit cell of AB_2O_4 there are 8 tetrahedral and 4 octahedral holes



Normal Spinel: A^{2+} ions occupy the T_d holes and both B^{3+} ions occupy the octahedral holes

Eg. $MgAl_2O_4$
Mg^{2+} Tetrahedral
Al^{3+} octahedral

Inverse Spinel: One B^{3+} ions occupy the T_d holes, and the (A^{2+} and one B^{3+}) ions occupy the octahedral holes

Eg. Fe_3O_4 (Magnetite)
Fe^{3+} Tetrahedral
Fe^{2+}, Fe^{3+} octahedral

Uses of Spinels:

- Manganese spinels used as the cathode in rechargeable batteries.
- Mixed-metal spinels, such as $ZnFe_2O_4$, have very useful magnetic properties.
- Several spinels—such as chromite, magnetite, and franklinite are important ores of metals

Question: Why does some AB_2O_4 compounds having transition elements as A and /or B prefer the inverse Spinel structure and some others normal Spinel structure?

ANS: Crystal Field Stabilization Energy in an octahedral site [Also given as Octahedral site stabilization energy (OSSE) in some books]

$Mn_3O_4 = Mn^{II}Mn^{III}{}_2O_4$

O^{2-} = a weak field ligand
Mn^{2+} = d^5 HS : CFSE = 0
Mn^{3+} = d^4 HS : CFSE = $-0.6\ \Delta_o$

Mn^{2+} by exchanging positions with Mn^{3+} in an octahedral hole is not going to gain any extra crystal field stabilization energy. While Mn^{3+} by being in the octahedral hole will have CFSE.
Therefore Mn_3O_4 will be Normal Spinel

Spinels or Inverse Spinels

$Fe_3O_4 = Fe^{II}Fe^{III}{}_2O_4$

$Fe^{2+} = d^6$ HS : CFSE =-0.4 Δ_o

$Fe^{3+} = d^5$ HS: CFSE = 0

Fe^{2+} by exchanging positions with Fe^{3+} to an octahedral hole is going to gain extra crystal field stabilization energy. While Fe^{3+} by being in the octahedral hole will not have any CFSE.

Therefore Fe_3O_4 will be Inverse Spinel

$(Fe^{III})^{tet}[Fe^{II}Fe^{III}]^{oct}O_4$

If A and B of AB_2O_4 are both s or p block elements (e.g. $CaAl_2O_4$), why it always show Spinel structure?

Home work

$Zn^{II}Fe^{III}_2O_4$

Zn^{2+} (d^{10}): CFSE = 0
Fe^{III} (d^5 high-spin): CFSE = 0

$Fe^{II}Cr^{III}_2O_4$

Fe^{2+} (d^6 high-spin): CFSE = $-0.4\Delta_O$
Cr^{III} (d^3 high-spin): CFSE = $-1.2\Delta_O$
Cr(III) has a greater CFSE compared to Fe(II);
Therefore, prefers to occupy octahedral site

Co_3O_4 is a normal spinel, whereas Fe_3O_4 (magnetite) is an inverse spinel. Explain.

$Ni^{II}Ga^{III}_2O_4$

Ni^{2+} (d^8 high-spin): CFSE = $-1.2\Delta_O$
Ga^{III} (d^{10}): CFSE = 0
Ni(II) has CFSE, therefore more stabilized in an octahedral site. Therefore the compound is inverse spinal: $(Ga^{III})^{tet}[Ni^{II}Ga^{III}]^{oct}O_4$

$Co^{II}Fe^{III}_2O_4$

Co^{2+} (d^7, high-spin): CFSE = $-0.8\Delta_O$
Fe^{III} (d^5, high-spin): CFSE = 0
Co(II) has CFSE, more stabilized in an octahedral site. Therefore the compound is inverse spinal: $(Fe^{III})^{tet}[Co^{II}Fe^{III}]^{oct}O_4$

Problem Solving

Following is the electronic absorption spectra of an aqueous solution of $CrCl_3{\cdot}6H_2O$ (solid line). It has been observed that the spectrum is blue shifted over time upon keeping the solution for hours as indicated by short dashes and dot-dashes. Explain the observation based on your knowledge about crystal field theory.

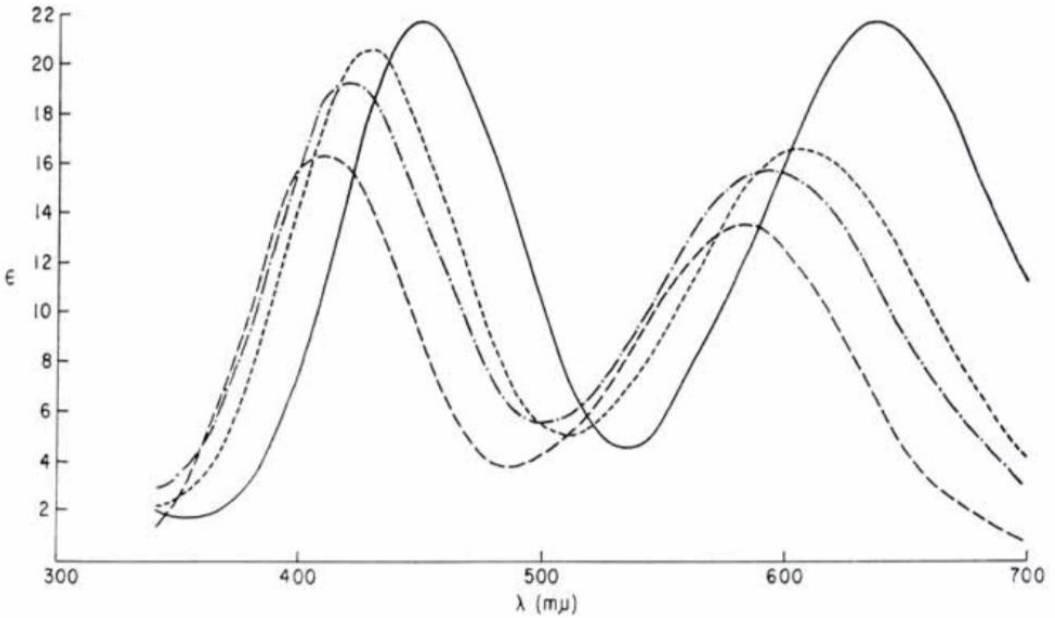
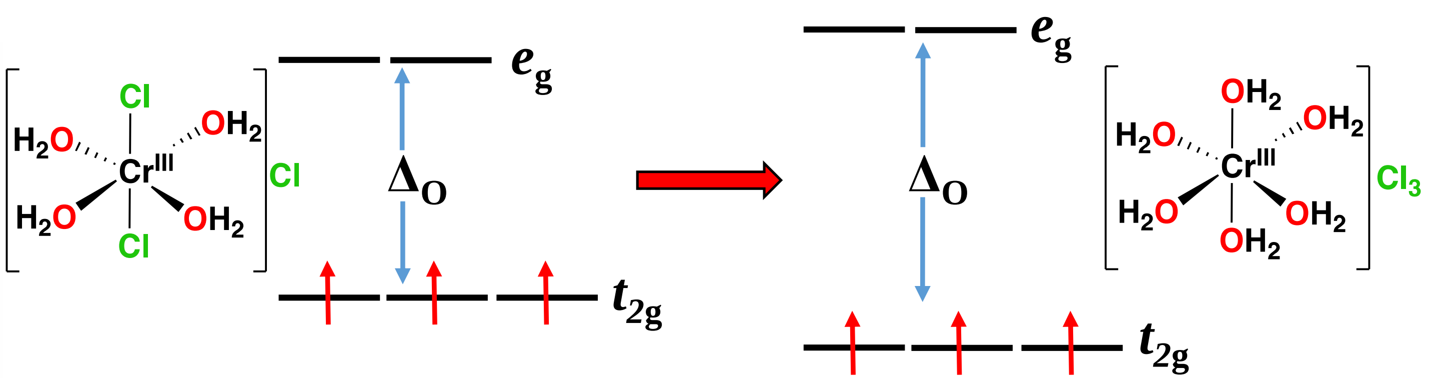
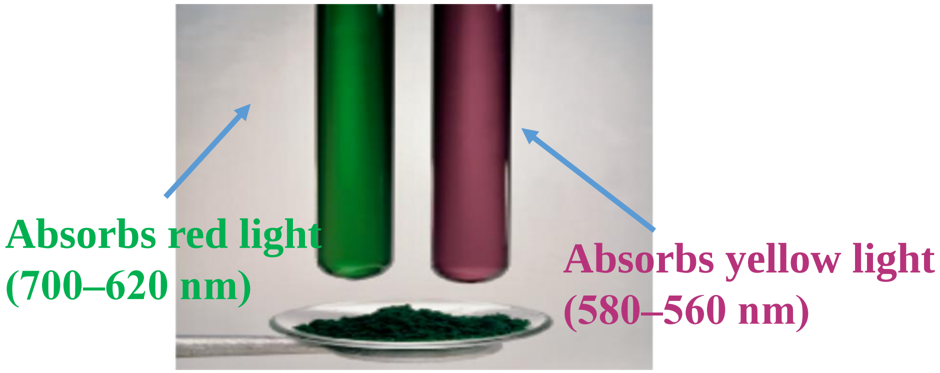


Figure 4. The solid line represents absorption spectrum of an aqueous solution of $CrCl_3{\cdot}6H_2O$ immediately after preparation; the short dashes represent the same solution after two hours; the dot-dash line represents the solution after 6.5 hours; and the long dashes represent the spectrum of $Cr(H_2O)_6(NO_3)_3{\cdot}3H_2O$.



Exchange of Cl^- by H_2O

Increase of Δ_O value



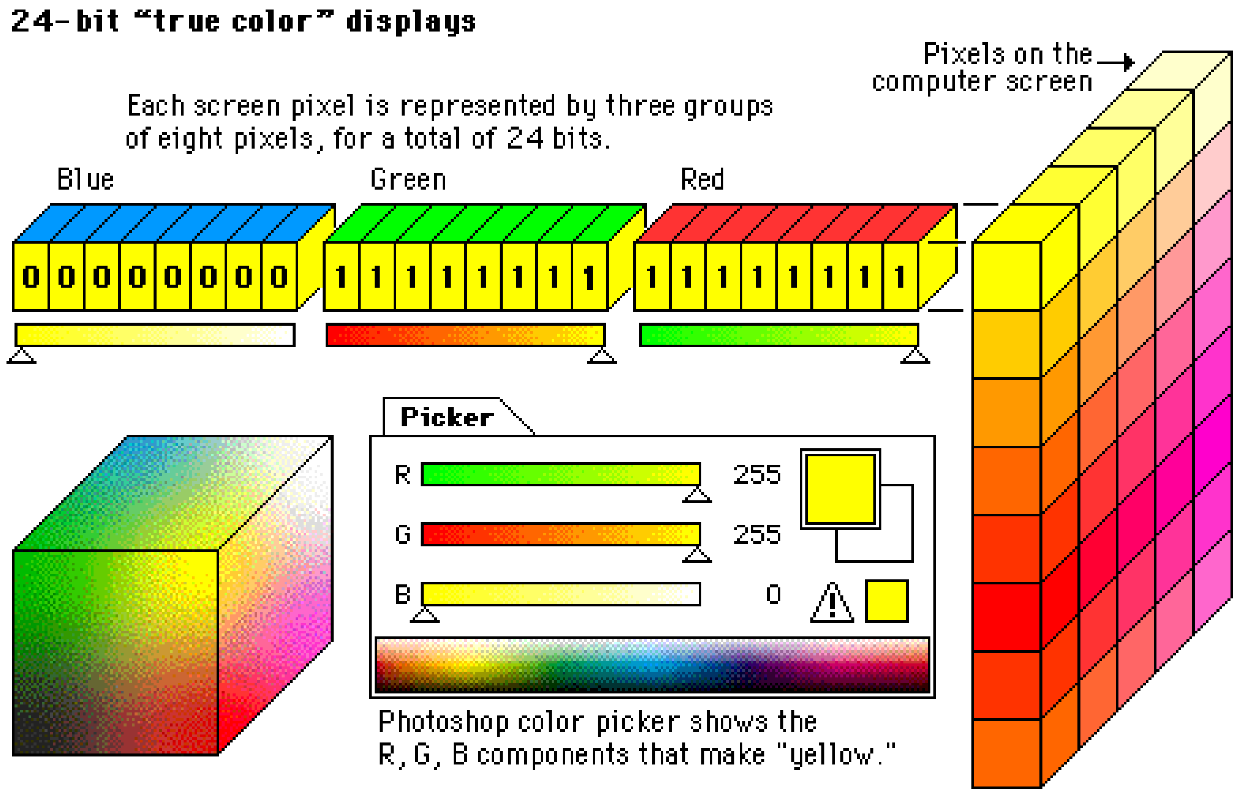
Change of the color of an aqueous solution of $CrCl_3$ with time

Color

BY CHRISTINA ROSSETTI 1830–1894

What is pink? a rose is pink
By a fountain's brink.
What is red? a poppy's red
In its barley bed.
What is blue? the sky is blue
Where the clouds float thro'.
What is white? a swan is white
Sailing in the light.
What is yellow? pears are yellow,
Rich and ripe and mellow.
What is green? the grass is green,
With small flowers between.
What is violet? clouds are violet
In the summer twilight.
What is orange? Why, an orange,
Just an orange!
Source: *The Golden Book of Poetry* (1947)

Factors to be explained
1. Frequency /Wavelength of color
2. Intensity of the color



Advantages and Disadvantages of Crystal Field Theory

Advantages over Valence Bond theory

1. Explains colors of complexes
2. Explains magnetic properties of complexes (without knowing hybridization) and temperature dependence of magnetic moments.
3. Classifies ligands as weak and strong
4. Explains anomalies in physical properties of metal complexes
5. Explains distortion in shape observed for some metal complexes

Disadvantages or drawbacks

1. *Evidences for the presence of covalent bonding (orbital overlap) in metal complexes have been disregarded.*
   e.g Does not explain why CO although neutral is a very strong ligand

2. *Cannot predict shape of complexes (since not based on hybridization)*
3. *Charge Transfer spectra not explained by CFT alone*

Thank you